NAT Gateway

1. Create VPC

vpc-0222c98d739cfd9c1 / dev-vpc Actions

Details Info

VPC ID vpc-0222c98d739cfd9c1	**State** Available	**Block Public Access** Off	**DNS hostnames** Disabled
DNS resolution Enabled	**Tenancy** default	**DHCP option set** dopt-01e704dbadad12e12	**Main route table** rtb-0808d4b2c483e6189
Main network ACL acl-050fb9477e48cab87	**Default VPC** No	**IPv4 CIDR** 10.0.0.0/16	**IPv6 pool** –
IPv6 CIDR (Network border group) –	**Network Address Usage metrics** Disabled	**Route 53 Resolver DNS Firewall rule groups** –	**Owner ID** 478217620043
Encryption control ID –	**Encryption control mode** –		

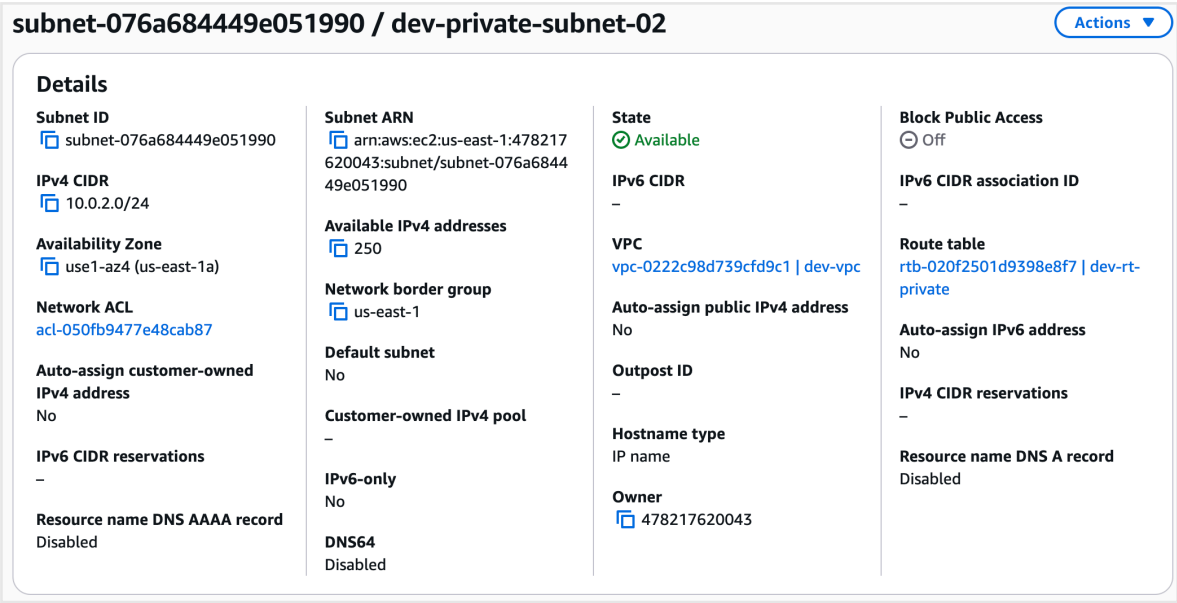
2. Create Public Subnet

subnet-063422b442164e43b / dev-public-subnet-01 Actions

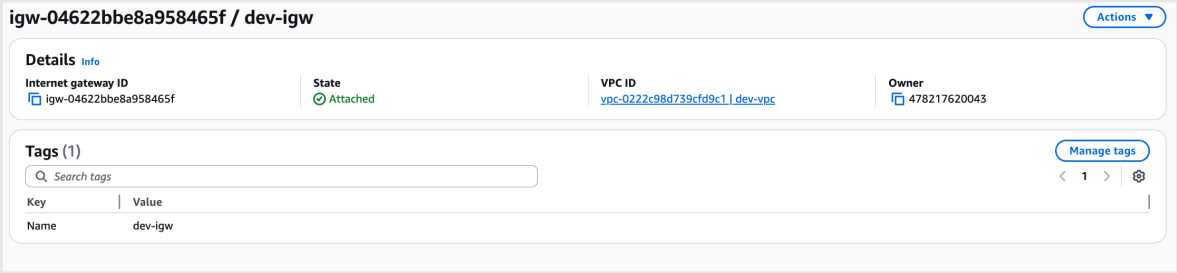
Details

Subnet ID subnet-063422b442164e43b	**Subnet ARN** arn:aws:ec2:us-east-1:478217620043:subnet/subnet-063422b442164e43b	**State** Available	**Block Public Access** Off
IPv4 CIDR 10.0.1.0/24	**Available IPv4 addresses** 249	**IPv6 CIDR** –	**IPv6 CIDR association ID** –
Availability Zone use1-az4 (us-east-1a)	**Network border group** us-east-1	**VPC** vpc-0222c98d739cfd9c1 | dev-vpc	**Route table** rtb-0d2b2c1e6e54ce296 | dev-rt-public
Network ACL acl-050fb9477e48cab87	**Default subnet** No	**Auto-assign public IPv4 address** No	**Auto-assign IPv6 address** No
Auto-assign customer-owned IPv4 address No	**Customer-owned IPv4 pool** –	**Outpost ID** –	**IPv4 CIDR reservations** –
IPv6 CIDR reservations –	**IPv6-only** No	**Hostname type** IP name	**Resource name DNS A record** Disabled
Resource name DNS AAAA record Disabled	**DNS64** Disabled	**Owner** 478217620043	

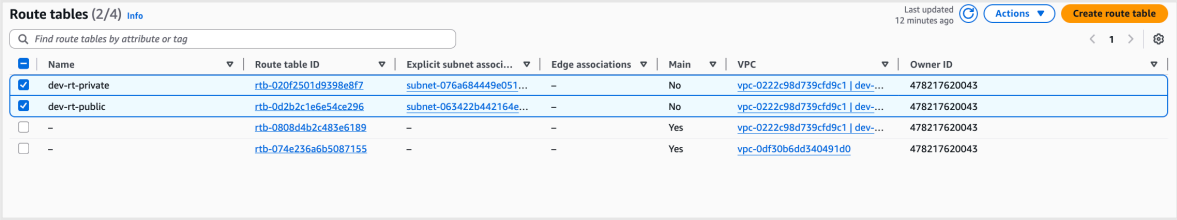
3. Create Private Subnet



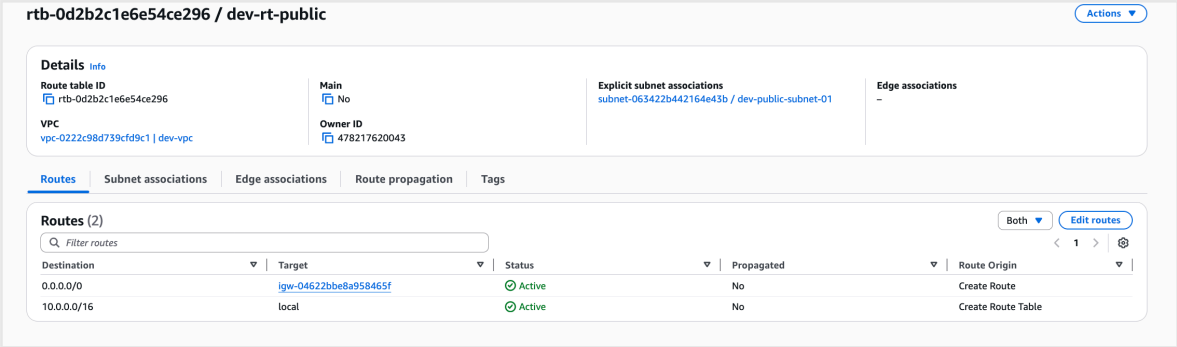
4. Create IGW



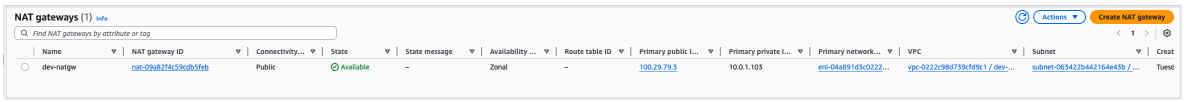
RT - Route Tables

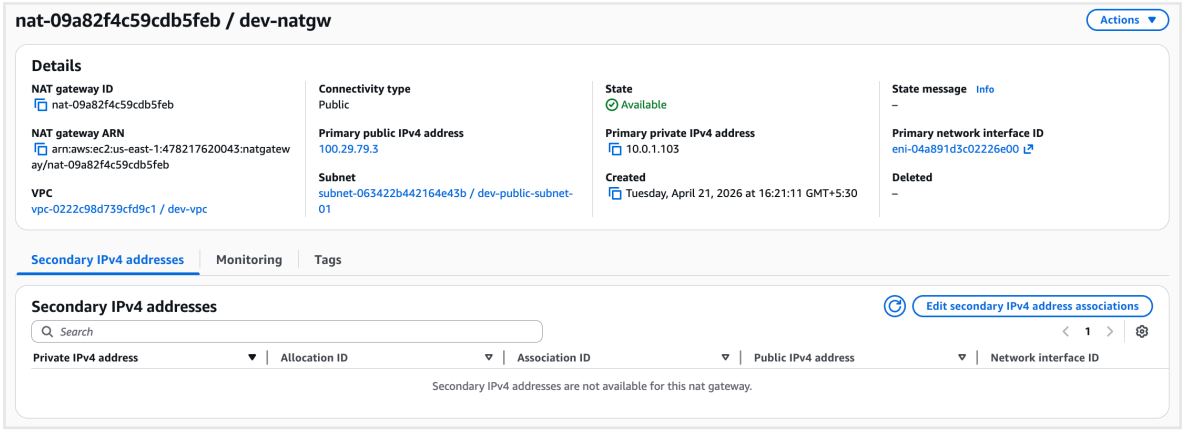


5. Create Public RT

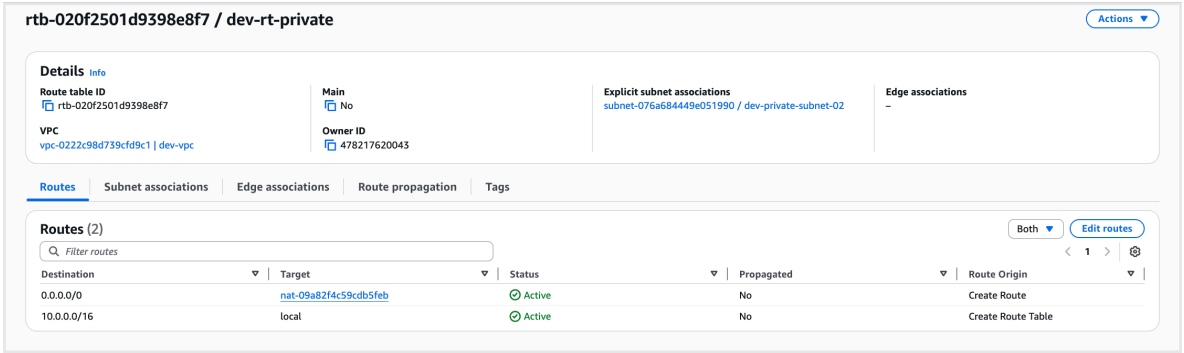


6. Create NAT

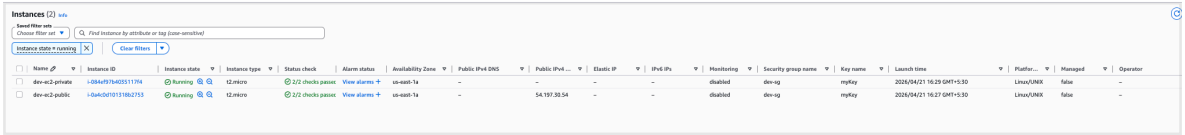




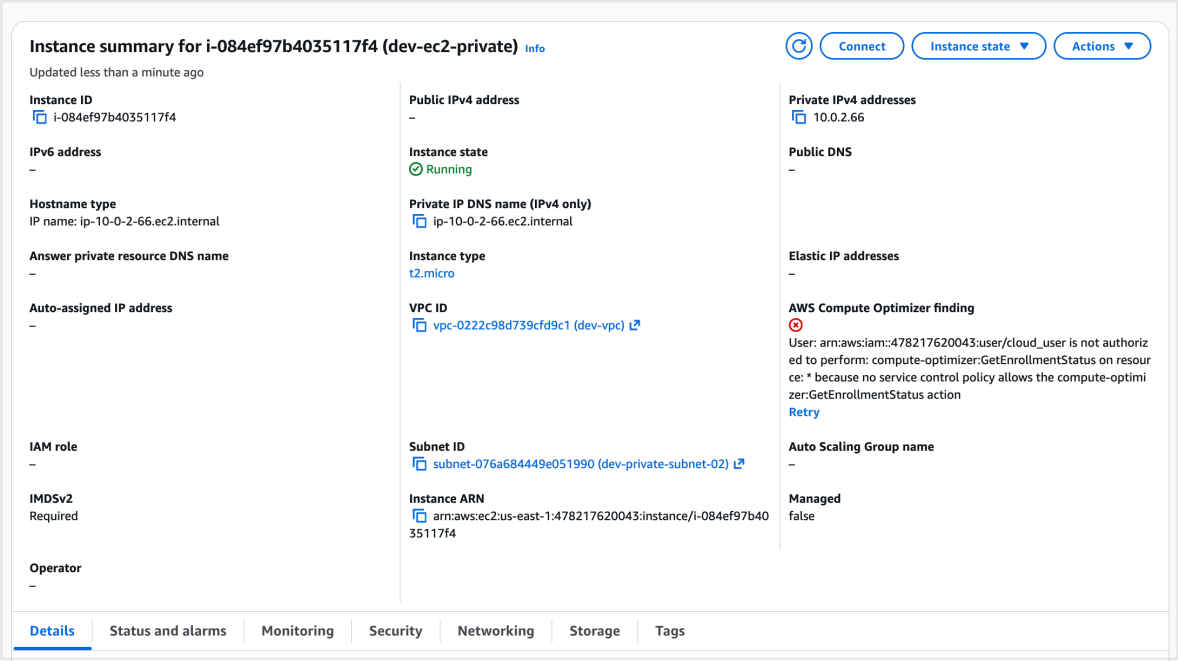
7. Create Private RT - NAT



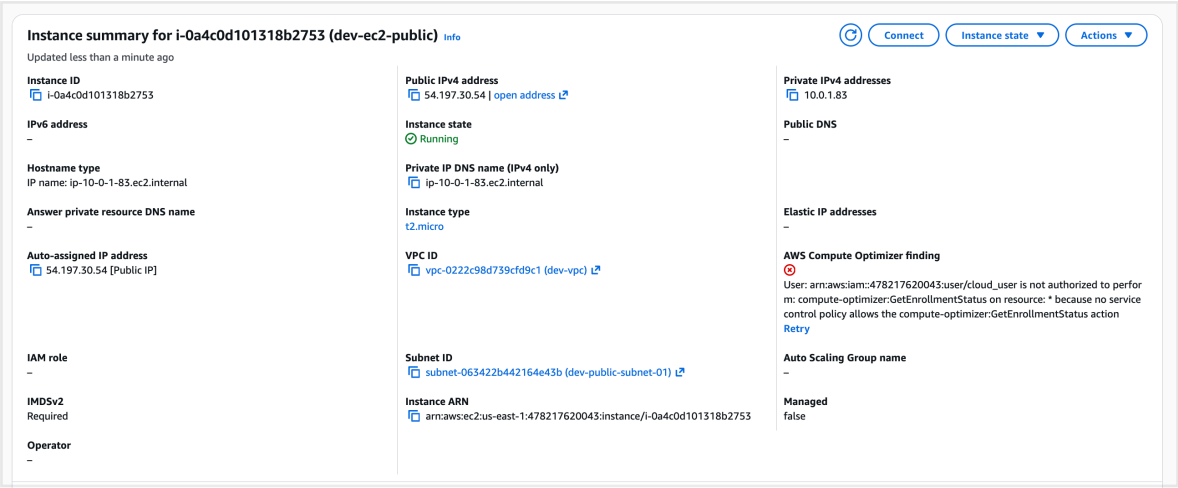
8. Create EC2 Instances



9. Create Private EC2 instance



10. Create Public EC2 instance



11. Now connect server via terminal...

```
Last login: Tue Apr 21 11:00:30 2026 from 223.185.40.8
[[ec2-user@ip-10-0-1-83 ~]$ touch myKey.pem
[[ec2-user@ip-10-0-1-83 ~]$ vi myKey.pem
[[ec2-user@ip-10-0-1-83 ~]$ vi myKey.pem
[[ec2-user@ip-10-0-1-83 ~]$ ls -l
total 4
-rw-r--r--. 1 ec2-user ec2-user 1680 Apr 21 11:06 myKey.pem
[[ec2-user@ip-10-0-1-83 ~]$ chmod 400 myKey.pem
[[ec2-user@ip-10-0-1-83 ~]$ ls -l
total 4
-r--------. 1 ec2-user ec2-user 1680 Apr 21 11:06 myKey.pem
[[ec2-user@ip-10-0-1-83 ~]$ ssh -i myKey.pem ec2-user@10.0.2.66
Load key "myKey.pem": error in libcrypto
ec2-user@10.0.2.66: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[[ec2-user@ip-10-0-1-83 ~]$ vi myKey.pem
[[ec2-user@ip-10-0-1-83 ~]$ ssh -i myKey.pem ec2-user@10.0.2.66
    ,     #_
    ~\_  ####_        Amazon Linux 2023
   ~~  \_#####\
   ~~     \###|
   ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
    ~~       V~' '->
     ~~~         /
       ~~._.   _/
          _/ _/
        _/m/'
[[ec2-user@ip-10-0-2-66 ~]$ ls
[[ec2-user@ip-10-0-2-66 ~]$ ping www.google.com
PING www.google.com (142.251.157.119) 56(84) bytes of data.
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=1 ttl=116 time=2.08 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=2 ttl=116 time=1.27 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=3 ttl=116 time=1.27 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=4 ttl=116 time=1.29 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=5 ttl=116 time=1.28 ms
```

```
ashishlanghe@Ashishs-MacBook-Pro ~ % ssh -i myKey.pem ec2-user@54.197.30.54
       ,     #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
Last login: Tue Apr 21 11:03:37 2026 from 223.185.40.8
[ec2-user@ip-10-0-1-83 ~]$ ssh -i myKey.pem ec2-user@10.0.2.66
       ,     #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
Last login: Tue Apr 21 11:08:53 2026 from 10.0.1.83
[ec2-user@ip-10-0-2-66 ~]$ ping www.google.com
PING www.google.com (142.251.156.119) 56(84) bytes of data.
64 bytes from 142.251.156.119 (142.251.156.119): icmp_seq=1 ttl=116 time=2.32 ms
64 bytes from 142.251.156.119 (142.251.156.119): icmp_seq=2 ttl=116 time=1.08 ms
64 bytes from 142.251.156.119 (142.251.156.119): icmp_seq=3 ttl=116 time=1.12 ms
64 bytes from 142.251.156.119 (142.251.156.119): icmp_seq=4 ttl=116 time=1.84 ms
64 bytes from 142.251.156.119 (142.251.156.119): icmp_seq=5 ttl=116 time=1.35 ms
64 bytes from 142.251.156.119 (142.251.156.119): icmp_seq=6 ttl=116 time=1.59 ms
64 bytes from 142.251.156.119 (142.251.156.119): icmp_seq=7 ttl=116 time=1.24 ms
64 bytes from 142.251.156.119 (142.251.156.119): icmp_seq=8 ttl=116 time=1.07 ms
64 bytes from 142.251.156.119 (142.251.156.119): icmp_seq=9 ttl=116 time=1.13 ms
^Z
[1]+  Stopped                 ping www.google.com
[ec2-user@ip-10-0-2-66 ~]$
```